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1. (a) Show that an appropriate integrating factor for

$$x^2 \frac{dy}{dx} + y = 1, \quad x > 0$$

is $e^{-1/x}$

[4]

- (b) Hence find the solution of the differential equation

$$x^2 \frac{dy}{dx} + y = 1, \quad x > 0$$

for which $y = 3$ when $x = 1$

[6]

2. (a) Show that

$$\frac{d}{dx}(\ln(\cosh x + 1)) = \tanh\left(\frac{x}{2}\right) \quad [3]$$

(b) Hence solve the differential equation

$$\frac{dy}{dx} + \tanh\left(\frac{x}{2}\right)y = \sinh x$$

for which $y = 2$ when $x = 0$. Give your answer in exact form [7]

- 3. (a)** For $x > -1$, determine the general solution of the differential equation

$$(1+x) \frac{dy}{dx} + y = \ln(1+x)$$

giving your answer in the form $y = f(x)$ [4]

- (b)** Given that $y = -1$ when $x = 0$, determine the positive value of x for which $y = 0$ [2]

4. (a) Show that, for $0 < x < \frac{\pi}{2}$, an appropriate integrating factor for

$$\cos x \frac{dy}{dx} + y \sin x = x$$

is $\sec x$

[4]

- (b) Hence find the solution of

$$\cos x \frac{dy}{dx} + y \sin x = x$$

for $0 < x < \frac{\pi}{2}$, given that $y = 1$ when $x = \frac{\pi}{4}$

[6]

5. (a) Show that, for $x > 0$,

$$\frac{d}{dx} \left(\ln \left(\coth \frac{x}{2} \right) \right) = -\operatorname{cosech} x \quad [3]$$

(b) Hence solve the differential equation

$$\frac{dy}{dx} - \operatorname{cosech} x y = \tanh \frac{x}{2}, \quad x > 0$$

for which $y = 1$ when $x = \ln 3$. Give your answer in exact form [7]

6. (a) Show that an appropriate integrating factor for

$$x(x-1) \frac{dy}{dx} + (x+1)y = x(x-1)^2$$

is $\frac{(x-1)^2}{x}$ [4]

(b) Hence find the solution of the differential equation

$$x(x-1) \frac{dy}{dx} + (x+1)y = x(x-1)^2$$

for $x > 1$, given that $y = 2$ when $x = 2$ [6]

7. Find the general solution of the differential equation

$$\frac{dy}{dx} + \cot x \, y = \frac{\cos x}{\sin^2 x (1 + \ln^2(\sin x))}$$

where $0 < x < \pi$

[7]

8. (a) Solve the differential equation

$$(1+x) \frac{dy}{dx} - 2y = (1+x)^4, \quad x > -1$$

giving your answer in the form $y = f(x)$ [3]

(b) The solution curve passes through the point $(0, -1)$. Find the exact positive value of x for which $y = 0$ [3]

9. (a) Show that

$$\frac{d}{dx} \ln(\sec x + \tan x) = \sec x \quad [2]$$

(b) Hence find the general solution of the differential equation

$$\frac{dy}{dx} - 2y \tan x = \sec^3 x$$

$$\text{where } -\frac{\pi}{2} < x < \frac{\pi}{2}. \quad [5]$$

10. (a) Show that

$$\frac{d}{dx} \left(\ln \left(\frac{\cosh x + 1}{\sinh x} \right) \right) = -\operatorname{cosech} x \quad [3]$$

(b) Hence solve, for $x > 0$,

$$\sinh x \frac{dy}{dx} - y = 1 - \cosh x$$

given that $y = 2$ when $x = \ln 3$. Give your answer in exact form [7]

- 11. (a)** Find the general solution of the differential equation

$$\frac{dy}{dx} + \frac{2x}{x^2 + 1}y = 6x$$

giving your answer in the form $y = f(x)$

[3]

- (b)** Given that $y = \frac{7}{2}$ when $x = 0$, find the smallest positive value of x for which $y = 4$

[3]

12. For $x > 0$, solve the differential equation

$$\sinh x \frac{dy}{dx} - 2y \cosh x = \sinh^2 x \cosh x$$

given that $y = 5$ when $x = \ln(1 + \sqrt{2})$.

Give your answer in exact form.

[7]

13. For $x > 0$, a function y satisfies

$$\sinh x \frac{dy}{dx} - y \cosh x = \sinh x \cosh^3 x$$

(a) By first writing the equation in the form $\frac{dy}{dx} + P(x)y = Q(x)$, find an integrating factor [2]

(b) Hence solve the differential equation, given that $y = 2$ when $x = \ln(1 + \sqrt{2})$ [5]

Give your answer in exact form.

14. For $x > 0$, solve the differential equation

$$\sinh x \frac{dy}{dx} - y \cosh x = \sinh x \cosh x$$

given that $y = 2$ when $x = \ln(1 + \sqrt{2})$.

Give your answer in exact form.

[7]

15. Find the general solution of the differential equation

$$\frac{dy}{dx} - y \tan x = \frac{1}{\sqrt{1 + \sin x}}$$

where $-\frac{\pi}{2} < x < \frac{\pi}{2}$

[7]